

Deflection

$$d\psi = \frac{ds}{R} = \frac{(dy^2 + dx^2)^{1/2}}{R}$$

$$d\psi = \frac{ds}{R} = \frac{dx \sqrt{1 + \left(\frac{dy}{dx}\right)^2}}{R}$$

$$\frac{d\psi}{dx} = \frac{\sqrt{1 + \left(\frac{dy}{dx}\right)^2}}{R} \quad (1)$$

$$\tan \psi = \frac{dy}{dx} \rightarrow \text{Diff}$$

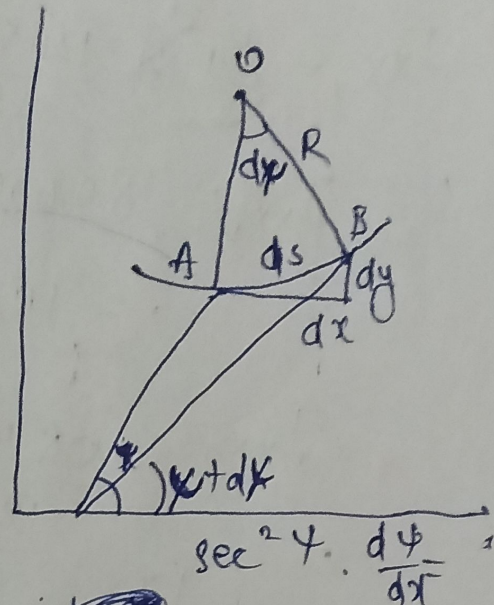
$$\sec^2 \psi = \frac{d^2y}{dx^2}$$

$$\frac{d\psi}{dx} = \frac{d^2y/dx^2}{1 + \left(\frac{dy}{dx}\right)^2}$$

$$\frac{\sqrt{1 + \left(\frac{dy}{dx}\right)^2}}{R} = \frac{d^2y}{dx^2} \left[1 + \left(\frac{dy}{dx}\right)^2 \right]^{3/2}$$

$$\frac{M}{EI} = \frac{d^2y}{dx^2} \left[1 + \left(\frac{dy}{dx}\right)^2 \right]^{3/2}$$

$$M = EI \frac{d^2y}{dx^2}$$



$$\sec^2 \psi = \frac{d^2y}{dx^2}$$

$$\frac{M}{EI} = \frac{1}{y} = \frac{E}{R}$$

$$EI y = wx(l-a) + 2wx(a-a)$$

$$= \frac{wx^2}{2} + \frac{2wx(a-a)^2}{2} + C_1$$

$$EI y = \frac{wx^3}{6} + \frac{wx(a-a)^3}{3} + C_1 x + C_2$$

$$x=l, \frac{dy}{dx} = 0$$

$$x=l, y = 0$$

$$-\frac{wl^2}{2} + \frac{wl(l-a)^2}{2} = C_1$$

$$C_2 = -\frac{wl^3}{6} + \frac{wl(l-a)^3}{3} + \frac{wl^2}{2} - \frac{wl(l-a)^2}{2}$$

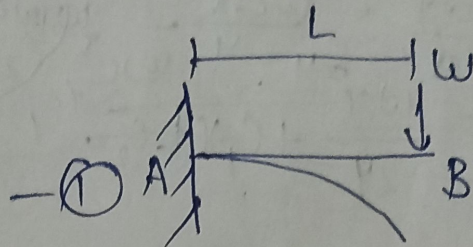
$$EI = \frac{wl^3}{3} + \frac{wl(l-a)^3}{3} - \frac{wl(l-a)^2}{2}$$

$$EI \frac{dy}{dx} = \int M dx + C_1$$

$$EI y = \iint M dx + C_1 x + C_2$$

$EI \rightarrow$ flexural rigidity

$$EI \frac{d^2 y}{dx^2} = wx$$



$$EI \frac{dy}{dx} = \frac{wx^2}{2} + C_1 \quad (2)$$

$$B_{BA} = wx$$

$$EI y = \frac{wx^3}{6} + C_1 x + C_2 \quad (3)$$

$$B_{BA} = \frac{wL^2}{2}$$

$$x=L, \frac{dy}{dx} = 0$$

$$x=L, y=0$$

from (2)

$$C_1 = -\frac{wL^2}{2}$$

from (3)

$$-\frac{wL^3}{6} + \frac{wL^3}{2} = C_2$$

$$\frac{wL^3}{2} \left[-\frac{1}{3} + 1 \right] = C_2$$

$$\frac{wL^3}{3} = C_2$$

Put $x=0$

$$y = \frac{1}{3} \frac{wL^3}{EI}$$

$$\left(\frac{1}{3} + 1 \right)$$

$$\frac{1+3}{3} \Rightarrow \frac{4}{3}$$

$$\left(-\frac{1}{6} + \frac{1}{2} \right) \frac{1+3}{3} = \frac{2}{3} = \frac{1}{3}$$

Deflection Double Integration method

$$EI \frac{d^2y}{dx^2} = M$$

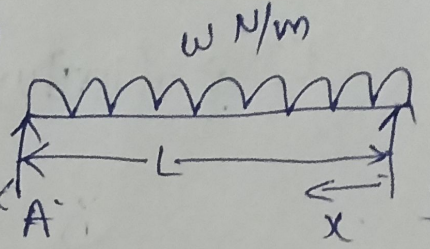
$$EI \frac{d^2y}{dx^2} = \frac{wLx}{2} - \frac{wx^2}{2}$$


Diagram of a simply supported beam of length L with a uniformly distributed load w N/m. The beam is supported at points A and B. The deflection curve is shown with a maximum deflection at the center. The load is represented by a wavy line above the beam.

$$EI \frac{dy}{dx} = \frac{wLx^2}{4} - \frac{wx^3}{6} + C_1$$

$$EI y = \frac{wLx^3}{12} - \frac{wx^4}{24} + C_1x + C_2$$

$$EI y = -\frac{wx^4}{24} + \frac{wLx^3}{12} + C_1x + C_2$$

$$x=0, y=0 \Rightarrow C_2=0$$

$$x=\frac{L}{2}, \frac{dy}{dx}=0$$

$$-\frac{wLx}{2} + \frac{wx^2}{2} = 0$$

$$\frac{wx^2}{2} = \frac{wLx}{2}$$

$$0 = \frac{wL^2}{4 \times 4} - \frac{wL^3}{6 \times 8} + C_1$$

$$C_1 = -\frac{wL^3}{16} + \frac{wL^3}{48}$$

$$C_1 = -\frac{wL^3}{24}$$

$$\text{Put } x = \frac{L}{2}$$

$$y = -\frac{wL^4}{24 \times 16} + \frac{wL^3}{12 \times 8} - \frac{wL^3 \times L}{24 \times 2}$$

$$EI$$

$$\begin{array}{r} \frac{x^2}{4} \\ \frac{x^3}{12} \\ 2 \times \frac{1}{16} \\ \hline 144 \\ 24 \times \\ \hline 304 \end{array}$$

$$y = \frac{-wL^4}{384} + \frac{wL^4}{96} - \frac{wL^4}{98}$$

$$EI$$

$$y = \frac{-wL^4 + 4wL^4 - 8wL^4}{384 \times EI}$$

$$= \frac{-5wL^4}{384 \times EI}$$

$$= \frac{-5wL^4}{384 \times EI}$$

$$\Rightarrow \frac{wL^4}{76.8 \times EI}$$

one

$$EI \frac{d^2y}{dx^2} = \frac{w}{2} x$$

$$EI \frac{dy}{dx} = \frac{w}{2} \frac{x^2}{2} + C_1$$

$$EI y = \frac{w}{24} x^3 + C_1 x + C_2$$

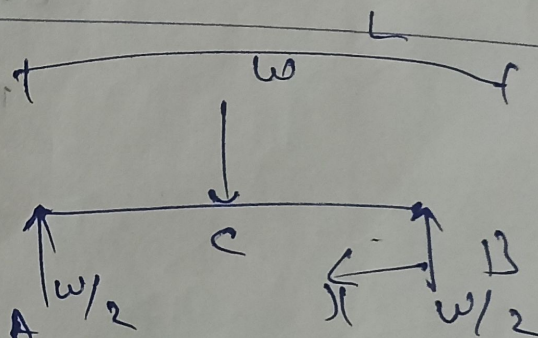
$$EI y = \frac{w}{12} x^3 + C_1 x + C_2$$

$$\frac{w}{2} \times \frac{L^2}{8} = 0$$

$$C_1 = -\frac{wL^2}{16}$$

$$y = \frac{wL^3}{12 \times 8} + \frac{wL^3}{16 \times 2}$$

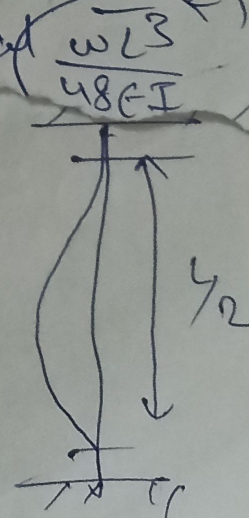
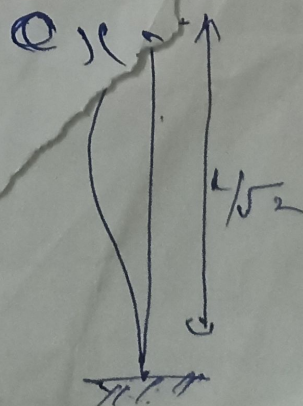
$$y = \frac{wL^3}{96} + \frac{wL^3}{32}$$



$$M_{BC} = \frac{w}{2} x$$

$$x = \frac{wL^2}{4 \times 16}$$

$$C_2 = 0 \quad \text{and} \quad \frac{wL^3}{48EI}$$



$$M_{BC} = \frac{w_a}{L} x$$

$$M_{CA} = \frac{w_a x}{L} - w(x-b)$$

Macaulay's

$$EI \frac{d^2 y}{dx^2} = \frac{w_a x}{L} - w(x-b)$$

$$EI \frac{dy}{dx} = \frac{w_a x^2}{2L} + C_1 - \frac{w(x-b)^2}{2}$$

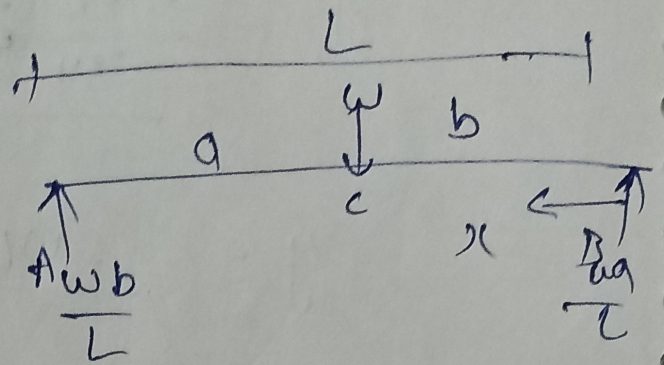
$$EI y = \frac{w_a x^3}{6L} + C_1 x + C_2 - \frac{w(x-b)^3}{6}$$

$$x=0, y=0$$

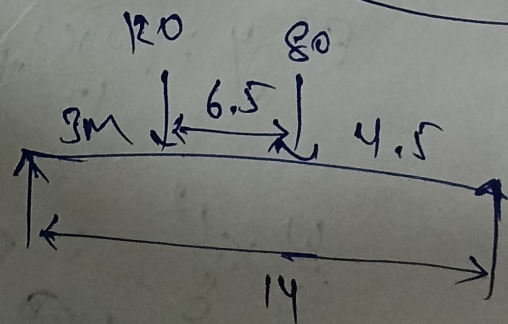
$$x=L, y=0$$

$$x=? \quad \frac{dy}{dx} = 0$$

$(x-b)$ always
Positive



11/11



$$y = \frac{24}{11}$$

Columns

$$EI \frac{d^2 y}{dx^2} = M$$

$$EI \frac{d^2 y}{dx^2} = -Py$$

$$\frac{d^2 y}{dx^2} + \frac{P}{EI} y = 0$$

$$y = C_1 \cos x \sqrt{\frac{P}{EI}} + C_2 \sin x \sqrt{\frac{P}{EI}}$$

$$x=0, y=0$$

$$0 = C_1 \sqrt{\frac{P}{EI}} + C_2 \times 0$$

$$C_1 = 0$$

$$x=L, y=0$$

$$0 = C_2 \sin L \sqrt{\frac{P}{EI}}$$

$$\sin L \sqrt{\frac{P}{EI}} = 0$$

$$L \sqrt{\frac{P}{EI}} = n\pi$$

$$P = \frac{\pi^2 E I}{L^2}$$

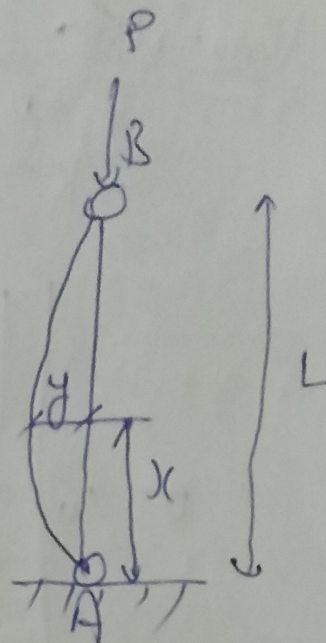
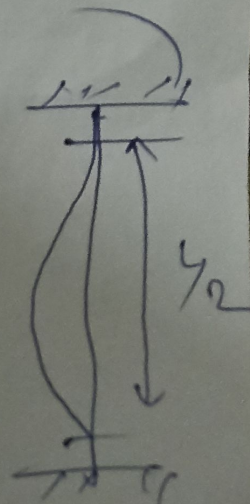
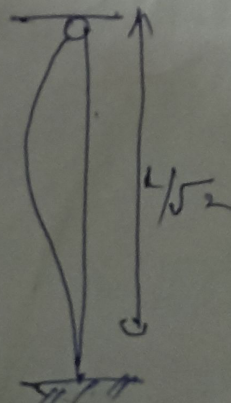
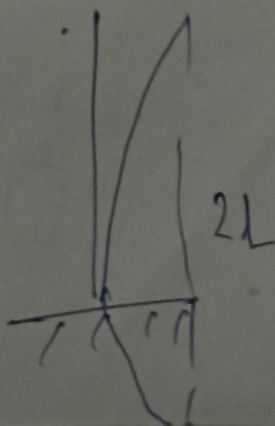
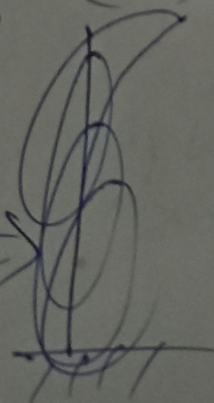
$$P = \frac{\pi^2 E I}{L^2}$$

Euler Load

$$\frac{P}{A} = \sigma = \frac{\pi^2 E}{(L/K)^2}$$

It is 1 so no need to write

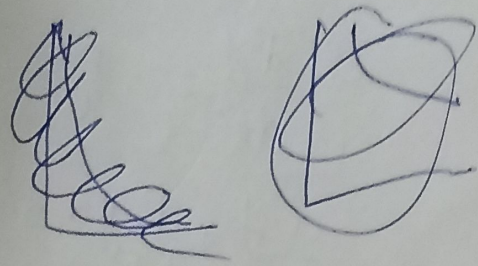
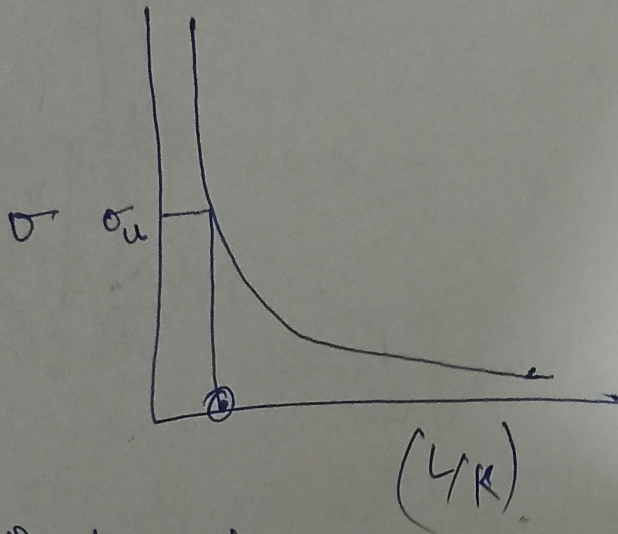
use $l = l_e$ for these



Limitation

$L/K = \text{Slenderness ratio}$

$$\frac{P}{A} = \sigma = \frac{\pi^2 E}{(L/K)^2}$$



Rankine formula

$$\frac{1}{P} = \frac{1}{P_c} + \frac{1}{P_E}$$

$$P_c = \sigma_c A$$

$$P_E = \frac{\pi^2 EI}{L^2}$$

Euler critical load

$$P = \frac{P_c P_E}{P_c + P_E}$$

$$= \frac{P_c}{1 + \frac{P_c}{P_E}} = \frac{\sigma_c A}{1 + \frac{\sigma_c A L^2}{\pi^2 E I K^2}}$$

$$\frac{P}{A} = \frac{\sigma_c}{1 + \left(\frac{\sigma_c}{\pi^2 E} \right) \left(\frac{L}{K} \right)^2}$$

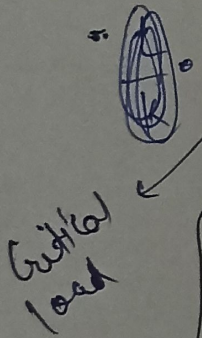
Rankine Constant for Column

$$\left(\frac{L}{K} \right)^2 = \frac{22 \times 2 \times 10^5}{7 \times 300}$$

$$\left(\frac{L}{K} \right)^2 = \frac{44 \times 10^3}{21}$$

$$L^2 = 2.09 \times 10^8$$

$$L = 81$$



Critical load

Crushing load

MOS

Principal stresses & Planes

$$\sigma_n = \frac{\sigma_1 + \sigma_2}{2} + \frac{\sigma_1 - \sigma_2}{2} \cos 2\theta + T \sin 2\theta$$

$$\sigma_t = \frac{\sigma_1 - \sigma_2}{2} \sin 2\theta - T \cos 2\theta$$

Condition (i):

Wt normal stress maximum

d/dθ σ_t = 0 (stress 0 at 90°)

σ_t = tangential stress

Put σ_t = 0

$$0 = \frac{\sigma_1 - \sigma_2}{2} \sin 2\theta - T \cos 2\theta$$

$$\tan 2\theta = \frac{2T}{\sigma_1 - \sigma_2}$$

$$\sigma_{np} = \frac{\sigma_1 + \sigma_2}{2} \pm \sqrt{\left(\frac{\sigma_1 - \sigma_2}{2}\right)^2 + T^2}$$

② σ_{t max} Different and Put

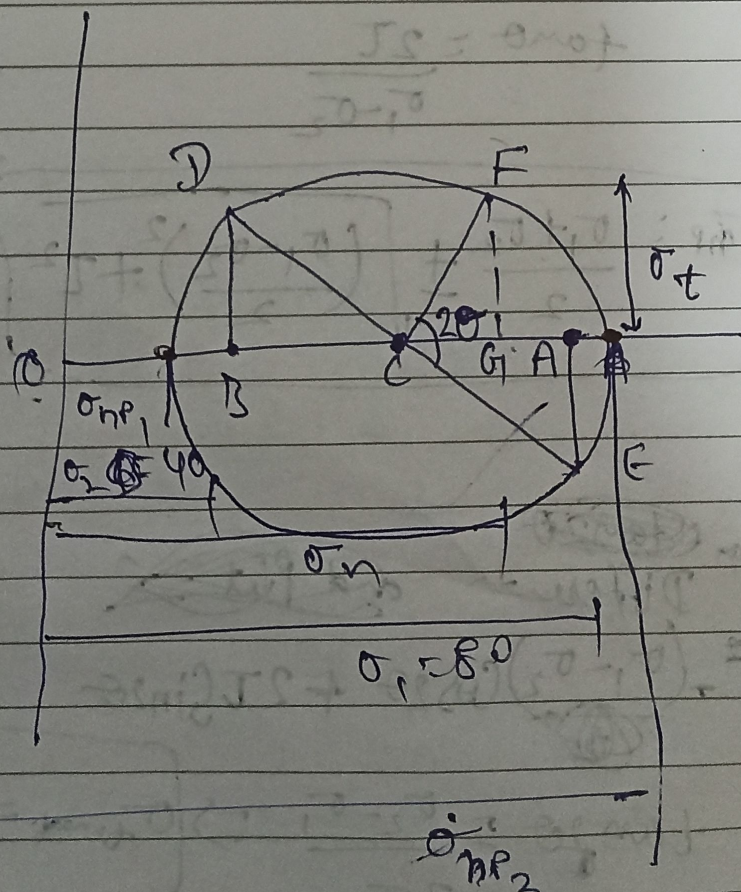
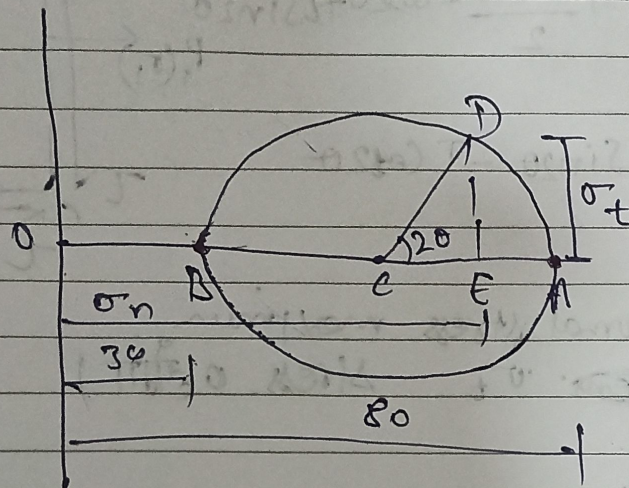
$$\frac{d\sigma_t}{d\theta} = (\sigma_1 - \sigma_2) \cos 2\theta + 2T \sin 2\theta$$

$$\tan 2\theta = \frac{\sigma_2 - \sigma_1}{2T}$$

$$\sigma_{t \max} = \sqrt{\left(\frac{\sigma_1 - \sigma_2}{2}\right)^2 + T^2}$$

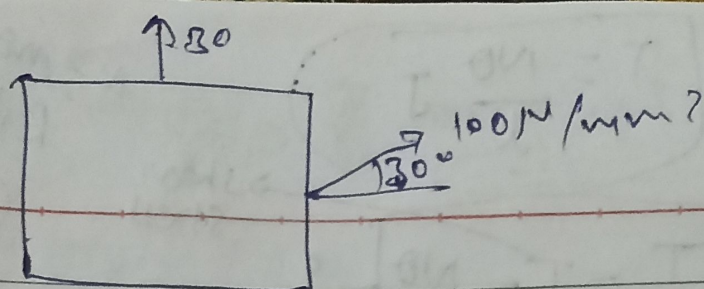
$$\sigma_{t \max} = \frac{\sigma_{np1} - \sigma_{np2}}{2}$$

Mohr's method :-



$\sigma_{t \max} = \text{radius}$

Ques



$$\sigma_1 = 100 \cos 30 \Rightarrow \frac{100\sqrt{3}}{2}$$

$$\sigma_2 = 30$$

$$\tau = 100 \sin 30$$

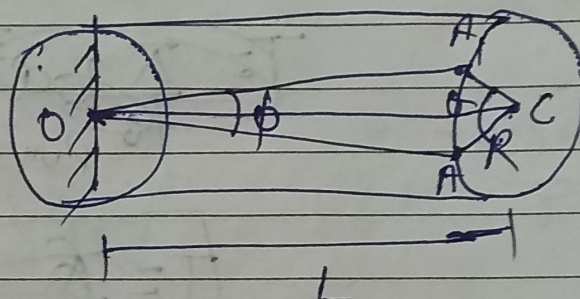
Torsion

$$\theta = \frac{AA_1}{R} \quad (1)$$

$$\phi = \frac{AA_1}{L}$$

$$\tau \propto \phi$$

$$\frac{\tau}{N} = \phi$$



$$AA_1 = \frac{\tau L}{N} \Rightarrow \theta = \frac{\tau L}{NR}$$

$$\boxed{\frac{\tau}{R} = \frac{N\theta}{L}}$$

If any arc is assumed from centre C.
then

$$\boxed{\frac{\tau}{R} = \frac{N\theta}{L}} = \frac{q}{r} \rightarrow \text{shear stress any distance}$$

Any variable stress

$$\delta T = q \cdot \delta a \cdot r$$

$$\delta T = \frac{Nq}{L} \cdot \delta a \cdot r^2$$

$$\Sigma \delta T = T = \frac{Nq}{L} \int \delta a \cdot r^2$$

$$T = \frac{NO}{\lambda} J$$

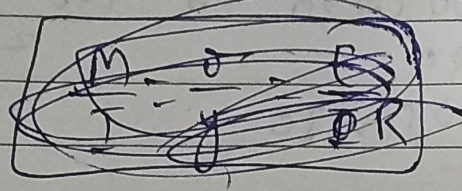
$N = \text{modulus of Rigidity}$

$$\frac{T}{J} = \frac{\tau}{R} = \frac{NO}{\lambda} = \frac{q}{\rho x}$$

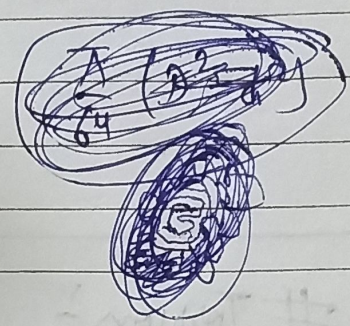
→ shear stress

$$J = I_{xx} + I_{yy}$$

Assumption write yourself



→ Extreme Radius



$$J = \frac{\pi}{64} d^4 + \frac{\pi}{64} d^4$$

$$R = \frac{d}{2}$$

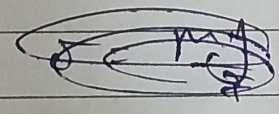
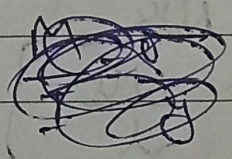
$$J = \frac{\pi}{16} d^3$$

$$\frac{T}{J} = \frac{\tau}{R}$$

$$T = \frac{\pi}{16} \tau d^3$$

Equivalent BM & TM

twisting moment



$$\sigma_p = \frac{\sigma}{2} + \sqrt{\left(\frac{\sigma}{2}\right)^2 + \tau^2}$$

$$T = \frac{\pi}{16} \tau d^3 \quad \Rightarrow \quad \frac{My}{I} = \frac{My}{\frac{\pi}{64} d^4} \quad \left| \quad T = \frac{\pi}{16} \tau d^3 \right.$$

$$\sigma_p = \frac{64 My}{\pi d^4} + \sqrt{\left(\frac{64 My}{\pi d^4} \times \frac{1}{2}\right)^2 + \frac{16 T}{\pi d^3}}$$

$$\sigma_p = \frac{\pi}{32} \sigma d^3 \quad (2)$$

~~$\tau = \frac{M}{J} \Rightarrow \tau \propto \frac{M}{d^3}$~~

~~$\tau = \frac{M}{J}$~~ M_e

$$\frac{\pi (\sigma_{np}) d^3}{32} = \frac{1}{2} [M + \sqrt{M^2 + T^2}]$$

$$\tau_{max} = \sqrt{\left(\frac{\sigma}{2}\right)^2 + T^2}$$

~~$\tau = \sqrt{M^2 + T^2} \tau_e$~~

lim $\frac{4 - \sin 3\pi}{5}$
 $x \rightarrow 0$

~~$\sqrt{\left(\frac{32M}{\pi d^3}\right)^2 + \left(\frac{16T}{\pi d^3}\right)^2}$~~

$\frac{4 - \sin 3\pi \times 3\pi}{5 - 3\pi \times 5}$

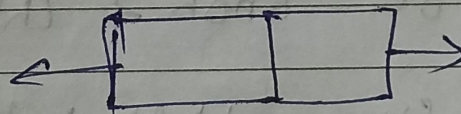
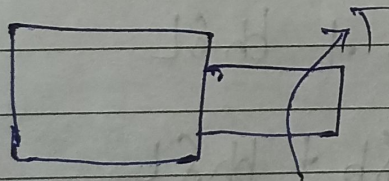
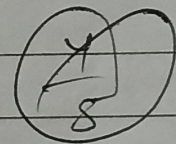
~~$\sqrt{\left(\frac{16M}{\pi d^3}\right)^2 + \left(\frac{16T}{\pi d^3}\right)^2}$~~

$\frac{4 - 3\pi}{5 - 5}$

$$= \sqrt{M^2 + T^2} \tau_e$$

$\frac{4 - 3\pi}{5}$

shafts in series & Parallel;

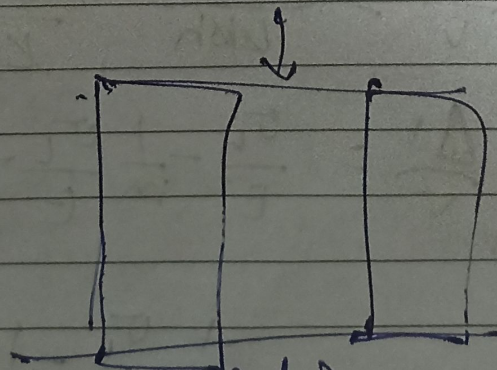


$$P = P_1 + P_2$$

$$T = T_1 = T_2$$

$$\delta = \delta_1 + \delta_2$$

$$\theta = \theta_1 + \theta_2$$

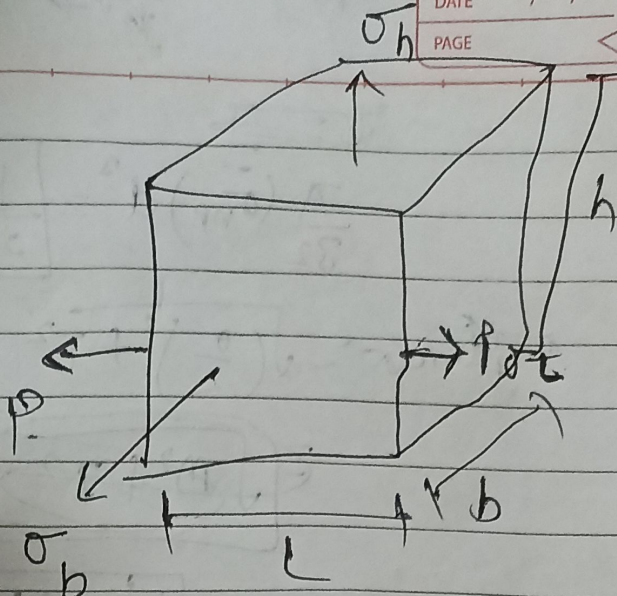


$$P = P_1 + P_2$$

$$\delta = \delta_1 + \delta_2$$

Three dimensional stress

Poisson's Ratio = $-\frac{\text{lateral strain}}{\text{long. strain}}$
 $(\mu) \left(\frac{1}{m}\right)$



$$\epsilon_L = \frac{\Delta L}{L} = \frac{\sigma_L}{E} - \frac{1}{m} \frac{\sigma_b}{E} - \frac{1}{m} \frac{\sigma_h}{E}$$

$$\epsilon_b = \frac{\Delta b}{b} = \frac{\sigma_b}{E} - \frac{1}{m} \frac{\sigma_L}{E} - \frac{1}{m} \frac{\sigma_h}{E}$$

$$\epsilon_h = \frac{\Delta h}{h} = \frac{\sigma_h}{E} - \frac{1}{m} \frac{\sigma_L}{E} - \frac{1}{m} \frac{\sigma_b}{E}$$

$$V = Lbh$$

$$\delta V = Lb\delta h + Lh\delta b + bh\delta L$$

$$\frac{\delta V}{V} = \frac{Lb\delta h + Lh\delta b + bh\delta L}{Lbh}$$

$$\frac{\delta V}{V} = \frac{Lb\delta h}{Lbh} + \frac{Lh\delta b}{Lbh} + \frac{bh\delta L}{Lbh}$$

$$\frac{\Delta V}{V} = \frac{\sigma_h}{E} - \frac{1}{m} \frac{\sigma_L}{E} - \frac{1}{m} \frac{\sigma_b}{E} + \frac{\sigma_b}{E} - \frac{1}{m} \frac{\sigma_L}{E} - \frac{1}{m} \frac{\sigma_h}{E} + \frac{\sigma_L}{E} - \frac{1}{m} \frac{\sigma_b}{E} - \frac{1}{m} \frac{\sigma_h}{E}$$

$$\frac{\Delta V}{V} = \frac{1}{E} \left[\sigma_h + \sigma_b + \sigma_L - \frac{3\sigma_L}{m} - \frac{3\sigma_b}{m} - \frac{3\sigma_h}{m} \right]$$

$$\frac{\Delta V}{V} = \frac{\sigma_L + \sigma_b + \sigma_h}{E} \left(1 - \frac{2}{m} \right)$$

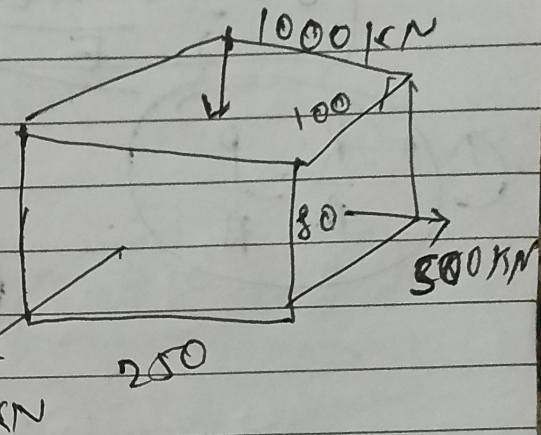
$$\frac{\Delta V}{V} = \left(\frac{\sigma_L + \sigma_b + \sigma_h}{E} \right) \left(1 - \frac{2}{m} \right)$$

numerical:- fig. show a block $250 \times 100 \times 80$ mm is subjected to axial load. 500 kN in tensile in 250 direction, 900 tensile in 250×80 mm face & 1000 kN compressive on 250×100 mm face.

Assuming Poisson's ratio $\mu = 0.25$

find the value of change in volume of the block

Young modulus $= 2 \times 10^5 \text{ N/mm}^2$
find change in all the three dimension?



$$\sigma_L = \frac{500 \times 10^3}{80000} \quad \sigma_b = \frac{900 \times 10^3}{250 \times 80} = 45$$

$$\sigma_h = \frac{1000 \times 10^3}{250 \times 100} = 40$$

$$\sigma_L = 62.5 \quad \sigma_b = 45 \quad \sigma_h = 40$$

$$\frac{\Delta V}{V} = \left(\frac{62.5 + 45 + 40}{2 \times 10^5} \right) \left(1 - 2 \times 0.25 \right)$$

$$337.5 \text{ mm}^3 \quad 73.75 \times 10^{-5} \times 5 \Rightarrow 36.87$$

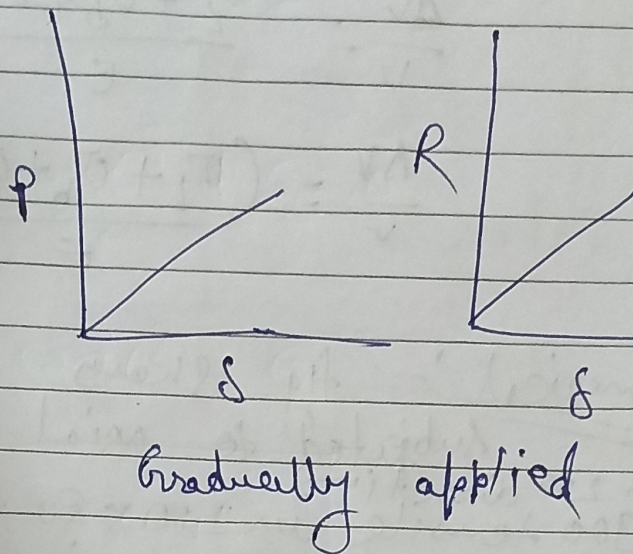
Strain Energy :-

$$\frac{1}{2} P \delta = \frac{1}{2} R \delta$$

$$R = P$$

$$U = \frac{1}{2} \sigma A \frac{\sigma L}{E}$$

$$U = \frac{1}{2} \frac{\sigma^2 A L}{E}$$

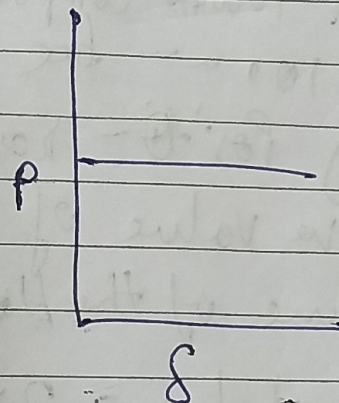


$$\sigma = \frac{R}{A} = \frac{P}{A}$$

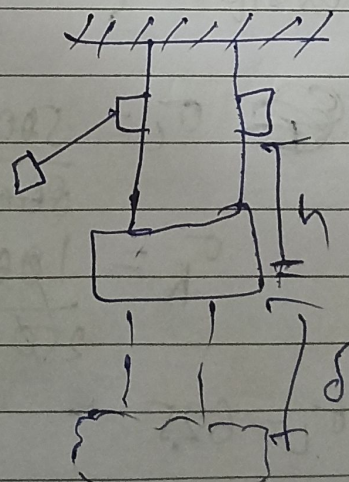
$$P \delta = \frac{1}{2} R \delta$$

$$R = 2P$$

$$\sigma = \frac{R}{A} = \frac{2P}{A}$$



$$P(h + \delta) = \frac{1}{2} \frac{\sigma^2 A L}{E}$$



Impact load

